because this reproduces the observed contrast dependence. A number of variants and extensions of this idea have also been considered, including, for example, that the denominator of this expression should include L factors for additional neurons with nearby receptive fields. This can account for the effects of visual stimuli outside the "classical" receptive field. Discussion of these effects is beyond the scope of this chapter.

2.5 Static Nonlinearities: Complex Cells

Recall that neurons in primary visual cortex are characterized as simple or complex. While linear methods, such as spike-triggered averages, are useful for revealing the properties of simple cells, at least to a first approximation, complex cells display features that are fundamentally incompatible with a linear description. The spatial receptive fields of complex cells cannot be divided into separate ON and OFF regions that sum linearly to generate the response. Areas where light and dark images excite the neuron overlap, making it difficult to measure and interpret spike-triggered average stimuli. Nevertheless, like simple cells, complex cells are selective to the spatial frequency and orientation of a grating. However, unlike simple cells, complex cells respond to bars of light or dark no matter where they are placed within the overall receptive field. Likewise, the responses of complex cells to grating stimuli show little dependence on spatial phase. Thus, a complex cell is selective for a particular type of image independent of its exact spatial position within the receptive field. This may represent an early stage in the visual processing that ultimately leads to position-invariant object recognition.

spatial-phase invariance

Complex cells also have temporal response characteristics that distinguish them from simple cells. Complex cell responses to moving gratings are approximately constant, not oscillatory as in figures 2.20 and 2.22. The firing rate of a complex cell responding to a counterphase grating oscillating with frequency ω has both a constant component and an oscillatory component with a frequency of 2ω, a phenomenon known as frequency doubling.

frequency doubling

Even though spike-triggered average stimuli and reverse-correlation functions fail to capture the response properties of complex cells, complex-cell responses can be described, to a first approximation, by a relatively straightforward extension of the reverse-correlation approach. The key observation comes from equation 2.34, which shows how the linear response estimate of a simple cell depends on spatial phase for an optimally oriented grating with K not too small. Consider two such responses, labeled L_1 and L_2, with preferred spatial phases ϕ and $\phi - \pi/2$. Including both the spatial and the temporal response factors, we find, for preferred spatial phase ϕ,

$$L_1 = AB(\omega, K)\cos(\phi - \Phi)\cos(\omega t - \delta)\,, \tag{2.39}$$

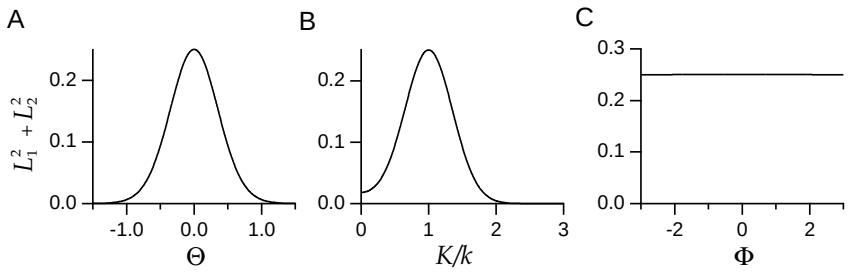

Figure 2.23 Selectivity of a complex cell model in response to a sinusoidal grating. The width and preferred spatial frequency of the Gabor functions underlying the estimated firing rate satisfy $k\sigma = 2$. (A) The complex cell response estimate, $L_1^2 + L_2^2$, as a function of stimulus orientation Θ for a grating with the preferred spatial frequency $K = k$. (B) $L_1^2 + L_2^2$ as a function of the ratio of the stimulus spatial frequency to its preferred value, K/k, for a grating oriented in the preferred direction $\Theta = 0$. (C) $L_1^2 + L_2^2$ as a function of stimulus spatial phase Φ for a grating with the preferred spatial frequency and orientation, $K = k$ and $\Theta = 0$.

where $B(\omega, K)$ is a temporal and spatial frequency-dependent amplitude factor. We do not need the explicit form of $B(\omega, K)$ here, but the reader is urged to derive it. For preferred spatial phase $\phi - \pi/2$,

$$L_2 = AB(\omega, K)\sin(\phi - \Phi)\cos(\omega t - \delta) \tag{2.40}$$

because $\cos(\phi - \pi/2 - \Phi) = \sin(\phi - \Phi)$. If we square and add these two terms, we obtain a result that does not depend on Φ,

$$L_1^2 + L_2^2 = A^2 B^2(\omega, K)\cos^2(\omega t - \delta)\,, \tag{2.41}$$

because $\cos^2(\phi - \Phi) + \sin^2(\phi - \Phi) = 1$. Thus, we can describe the spatial-phase-invariant response of a complex cell by writing

$$\mathsf{r}(t) = \mathsf{r}_0 + G\left(L_1^2 + L_2^2\right)\,, \tag{2.42}$$

for some constant G. The selectivities of such a response estimate to grating orientation, spatial frequency, and spatial phase are shown in figure 2.23. The response of the model complex cell is tuned to orientation and spatial frequency, but the spatial phase dependence, illustrated for a simple cell in figure 2.15C, is absent. In computing the curve for figure 2.23C, we used the exact expressions for L_1 and L_2 from the integrals in equations 2.31 and 2.32, not the approximation used in equation 2.34 to simplify the previous discussion. Although it is not visible in the figure, there is a weak dependence on Φ when the exact expressions are used.

The complex cell response given by equations 2.41 and 2.42 reproduces the frequency-doubling effect seen in complex cell responses because the factor $\cos^2(\omega t - \delta)$ oscillates with frequency 2ω. This follows from the identity

$$\cos^2(\omega t - \delta) = \frac{1}{2}\cos\left(2(\omega t - \delta)\right) + \frac{1}{2}\,. \tag{2.43}$$

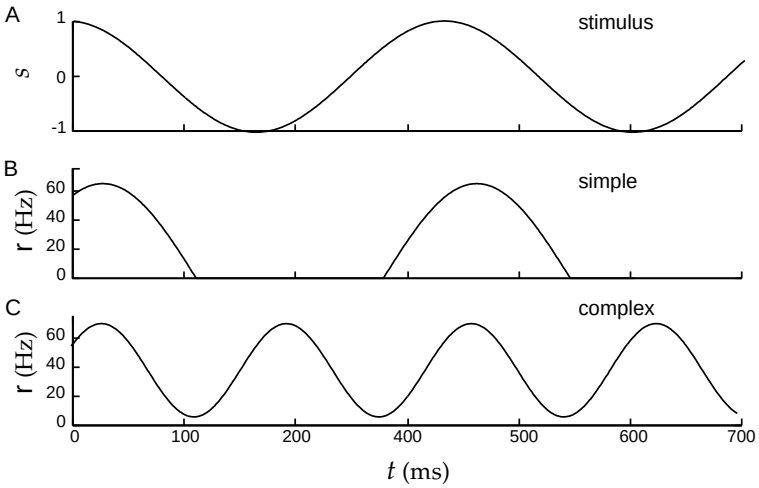

Figure 2.24 Temporal responses of model simple and complex cells to a counterphase grating. (A) The stimulus $s(x, y, t)$ at a given point (x, y) plotted as a function of time. (B) The rectified linear response estimate of a model simple cell to this grating with a temporal kernel given by equation 2.29 with $\alpha = 1/(15 \text{ ms})$. (C) The frequency-doubled response of a model complex cell with the same temporal kernel but with the estimated rate given by a squaring operation rather than rectification. The background firing rate is $r_0 = 5$ Hz. Note the temporal phase shift of both B and C relative to A.

In addition, the last term on the right side of this equation generates the constant component of the complex cell response to a counterphase grating. Figure 2.24 shows a comparison of model simple and complex cell responses to a counterphase grating, and illustrates this phenomenon.

energy model

The description of a complex cell response that we have presented is called an "energy" model because of its resemblance to the equation for the energy of a simple harmonic oscillator. The pair of linear filters used, with preferred spatial phases separated by $\pi/2$, is called a quadrature pair. Because of rectification, the terms L_1^2 and L_2^2 cannot be constructed by squaring the outputs of single simple cells. However, they can each be constructed by summing the squares of rectified outputs from two simple cells with preferred spatial phases separated by π. Thus, we can write the complex cell response as the sum of the squares of four rectified simple cell responses,

$$r(t) = r_0 + G\left([L_1]_+^2 + [L_2]_+^2 + [L_3]_+^2 + [L_4]_+^2\right) , \tag{2.44}$$

where the different $[L]_+$ terms represent the responses of simple cells with preferred spatial phases ϕ, $\phi + \pi/2$, $\phi + \pi$, and $\phi + 3\pi/2$. While such a construction is possible, it should not be interpreted too literally because complex cells receive input from many sources, including the LGN and other complex cells. Rather, this model should be viewed as purely descriptive. Mechanistic models of complex cells are described at the end of this chapter and in chapter 7.

2.6 Receptive Fields in the Retina and LGN

We end this discussion of the visual system by returning to the initial stages of the visual pathway and briefly describing the receptive field properties of neurons in the retina and LGN. Retinal ganglion cells display a wide variety of response characteristics, including nonlinear and direction-selective responses. However, a class of retinal ganglion cells (X cells in the cat or P cells in the monkey retina and LGN) can be described by a linear model built using reverse-correlation methods. The receptive fields of this class of retinal ganglion cells and an analogous type of LGN relay neurons are similar, so we do not treat them separately. The spatial structure of the receptive fields of these neurons has a center-surround structure consisting either of a circular central ON region surrounded by an annular OFF region, or the opposite arrangement of a central OFF region surrounded by an ON region. Such receptive fields are called ON-center and OFF-center, respectively. Figure 2.25A shows the spatial receptive fields of an ON-center cat LGN neuron.

The spatial structure of retinal ganglion and LGN receptive fields is well captured by a difference-of-Gaussians model in which the spatial receptive field is expressed as

difference of Gaussians

$$D_{\mathrm{s}}(x, y) = \pm\left(\frac{1}{2\pi\sigma_{\mathrm{cen}}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{\mathrm{cen}}^2}\right) - \frac{B}{2\pi\sigma_{\mathrm{sur}}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{\mathrm{sur}}^2}\right)\right). \tag{2.45}$$

Here the center of the receptive field has been placed at $x = y = 0$. The first Gaussian function in equation 2.45 describes the center, and the second, the surround. The size of the central region is determined by the parameter σ_{cen}, while σ_{sur}, which is greater than σ_{cen}, determines the size of the surround. B controls the balance between center and surround contributions. The $\pm$ sign allows both ON-center (+) and OFF-center (−) cases to be represented. Figure 2.25B shows a spatial receptive field formed from the difference of two Gaussians that approximates the receptive field structure in figure 2.25A.

Figure 2.25C shows that the spatial structure of the receptive field reverses over time with, in this case, a central ON region reversing to an OFF region as τ increases. Similarly, the OFF surround region changes to an ON region with increasing τ, although the reversal and the onset are slower for the surround than for the central region. Because of the difference between the time course of the center and of the surround regions, the space-time receptive field is not separable, although the center and surround components are individually separable. The basic features of LGN neuron space-time receptive fields are captured by

$$D(x, y, \tau) = \pm\left(\frac{D_{\mathrm{t}}^{\mathrm{cen}}(\tau)}{2\pi\sigma_{\mathrm{cen}}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{\mathrm{cen}}^2}\right) - \frac{BD_{\mathrm{t}}^{\mathrm{sur}}(\tau)}{2\pi\sigma_{\mathrm{sur}}^2}\exp\left(-\frac{x^2+y^2}{2\sigma_{\mathrm{sur}}^2}\right)\right). \tag{2.46}$$

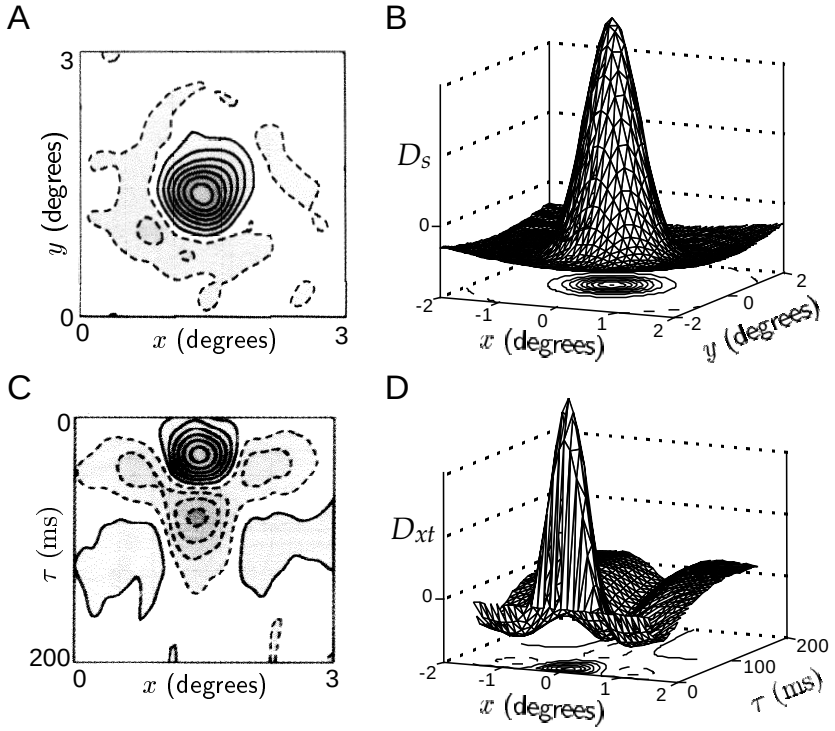

Figure 2.25 Receptive fields of LGN neurons. (A) The center-surround spatial structure of the receptive field of a cat LGN X cell. This has a central ON region (solid contours) and a surrounding OFF region (dashed contours). (B) A fit of the receptive field shown in A using a difference-of-Gaussians function (equation 2.45) with $\sigma_{\text{cen}} = 0.3°$, $\sigma_{\text{sur}} = 1.5°$, and $B = 5$. (C) The space-time receptive field of a cat LGN X cell. Note that the center and surround regions both reverse sign as a function of τ and that the temporal evolution is slower for the surround than for the center. (D) A fit of the space-time receptive field in C using equation 2.46 with the same parameters for the Gaussian functions as in B, and temporal factors given by equation 2.47 with $1/\alpha_{\text{cen}} = 16$ ms for the center, $1/\alpha_{\text{sur}} = 32$ ms for the surround, and $1/\beta_{\text{cen}} = 1/\beta_{\text{sur}} = 64$ ms. (A and C adapted from DeAngelis et al., 1995.)

Separate functions of time multiply the center and surround, but they can both be described by the same functions, using two sets of parameters,

$$D_{\text{t}}^{\text{cen,sur}}(\tau) = \alpha_{\text{cen,sur}}^2 \tau \exp(-\alpha_{\text{cen,sur}} \tau) - \beta_{\text{cen,sur}}^2 \tau \exp(-\beta_{\text{cen,sur}} \tau) . \quad (2.47)$$

The parameters α_{cen} and α_{sur} control the latency of the response in the center and surround regions, respectively, and β_{cen} and β_{sur} affect the time of the reversal. This function has characteristics similar to the function in equation 2.29, but the latency effect is less pronounced. Figure 2.25D shows the space-time receptive field of equation 2.46 with parameters chosen to match figure 2.25C.

Figure 2.26 shows the results of a direct test of a reverse-correlation model of an LGN neuron. The kernel needed to describe a particular LGN cell was first extracted by using a white-noise stimulus. This, together with

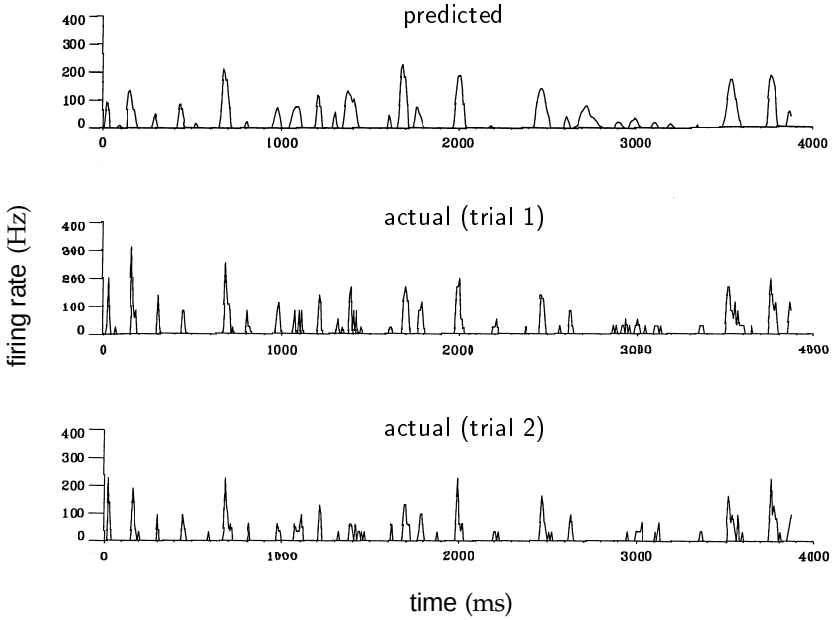

Figure 2.26 Comparison of predicted and measured firing rates for a cat LGN neuron responding to a video movie. The top panel is the rate predicted by integrating the product of the video image intensity and a linear filter obtained for this neuron from a spike-triggered average of a white-noise stimulus. The resulting linear prediction was rectified. The middle and lower panels are measured firing rates extracted from two different sets of trials. (Adapted from Dan et al., 1996.)

a rectifying static nonlinearity, was used to predict the firing rate of the neuron in response to a video movie. The top panel in figure 2.26 shows the resulting prediction, and the middle and lower panels show the actual firing rates extracted from two different groups of trials. The correlation coefficient between the predicted and actual firing rates was 0.5, which was very close to the correlation coefficient between firing rates extracted from different groups of trials. This means that the error of the prediction was no worse than the variability of the neural response itself.

2.7 Constructing V1 Receptive Fields

The models of visual receptive fields we have been discussing are purely descriptive, but they provide an important framework for studying how the circuits of the retina, LGN, and primary visual cortex generate neural responses. In an example of a more mechanistic model, Hubel and Wiesel (1962) proposed that the oriented receptive fields of cortical neurons could be generated by summing the input from appropriately selected LGN neurons. Their construction, shown in figure 2.27A, consists of alternating rows of ON-center and OFF-center LGN cells providing convergent input to a cortical simple cell. The left side of figure 2.27A shows the spatial ar-

Hubel-Wiesel simple cell model

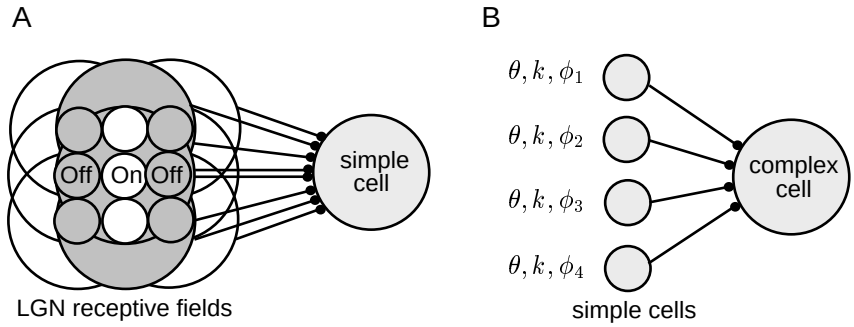

Figure 2.27 (A) The Hubel-Wiesel model of orientation selectivity. The spatial arrangement of the receptive fields of nine LGN neurons are shown, with a row of three ON-center fields flanked on either side by rows of three OFF-center fields. White areas denote ON fields and gray areas, OFF fields. In the model, the converging LGN inputs are summed by the simple cell. This arrangement produces a receptive field oriented in the vertical direction. (B) The Hubel-Wiesel model of a complex cell. Inputs from a number of simple cells with similar orientation and spatial frequency preferences (θ and k), but different spatial phase preferences (ϕ_1, ϕ_2, ϕ_3, and ϕ_4), converge on a complex cell and are summed. This produces a complex cell output that is selective for orientation and spatial frequency, but not for spatial phase. The figure shows four simple cells converging on a complex cell, but additional simple cells can be included to give a more complete coverage of spatial phase.

rangement of LGN receptive fields that, when summed, form bands of ON and OFF regions resembling the receptive field of an oriented simple cell. This model accounts for the selectivity of a simple cell purely on the basis of feedforward input from the LGN. We leave the study of this model as an exercise for the reader. Other models, which we discuss in chapter 7, include the effects of recurrent intracortical connections as well.

In a previous section, we showed how the properties of complex cell responses could be accounted for by using a squaring static nonlinearity. While this provides a good description of complex cells, there is little indication that complex cells actually square their inputs. Models of complex cells can be constructed without introducing a squaring nonlinearity. One such example is another model proposed by Hubel and Wiesel (1962), which is depicted in figure 2.27B. Here the phase-invariant response of a complex cell is produced by summing together the responses of several simple cells with similar orientation and spatial frequency tuning, but different preferred spatial phases. In this model, the complex cell inherits its orientation and spatial frequency preference from the simple cells that drive it, but spatial phase selectivity is reduced because the outputs of simple cells with a variety of spatial phase selectivities are summed. Analysis of this model is left as an exercise. While the model generates complex cell responses, there are indications that complex cells in primary visual cortex are not driven exclusively by simple cell input. An alternative model is considered in chapter 7.

Hubel-Wiesel complex cell model